

Yuan Xiao

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Education

Nanjing University <i>Ph.D., Software Engineering, Advisors: Chunrong Fang, Zhenyu Chen</i> <i>Dissertation submitted; defense expected in August 2026</i>	Sep. 2021 – Expected Sep. 2026 <i>Nanjing, Jiangsu, China</i>
Sun Yat-sen University <i>B.Sc., Mathematics and Applied Mathematics, Advisor: Xianping Guo</i> <i>Minor in Finance</i>	Sep. 2017 – Aug. 2021 <i>Guangzhou, Guangdong, China</i>

Research Experience

Visiting Ph.D. Student <i>School of Computing, National University of Singapore, Host: Jinsong Dong</i>	Nov. 2024 – Nov. 2025 <i>Singapore</i>
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Research Interests and Statement

Research Interests

Artificial intelligence; machine learning; trustworthy AI; AI security and safety.

Research Statement

My doctoral research centers on how training data shapes model behavior, using code dataset watermarking, data poisoning, and neural network robustness verification as the main technical vehicles to study how model behaviors are formed, verified, retained, and removed. This work has been published at CVPR, ISSTA, FSE, ACL Findings, TSE, and other international venues. Going forward, I plan to further investigate how training data drives the formation and evolution of model behavior, exploring the persistence, forgettability, and controllable intervention of model behaviors, in order to provide theoretical and technical support for trustworthy large models and intelligent software engineering.

Publications (* indicates co-first authorship)

Note: CCF A refers to the top-tier international computer science conferences and journals recommended by CCF (China Computer Federation); CORE Ranking is a widely used international grading system for computer science venues, classifying conferences into A*, A, B, and C tiers.

1. [First Author | CVPR 2024 | CCF A, Core Ranking A*]
[Towards General Robustness Verification of MaxPool-Based Convolutional Neural Networks via Tightening Linear Approximation.](#)
Yuan Xiao, Shiqing Ma, Juan Zhai, Chunrong Fang, Jinyuan Jia, Zhenyu Chen. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024: 24766–24775.
2. [First Author | CVPR 2025 | CCF A, Core Ranking A*]
[Tightening Robustness Verification of MaxPool-based Neural Networks via Minimizing the Over-Approximation Zone.](#)

- Yuan Xiao, Yuchen Chen, Shiqing Ma, Chunrong Fang, Tongtong Bai, Mingzheng Gu, Yuxin Cheng, Yanwei Chen, Zhenyu Chen. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025: 20695–20705.
3. [First Author | ISSTA 2025 | CCF A, Core Ranking A]
[DeCoMa: Detecting and Purifying Code Dataset Watermarks through Dual Channel Code Abstraction.](#)
Yuan Xiao, Yuchen Chen, Shiqing Ma, Haocheng Huang, Chunrong Fang, Yanwei Chen, Weisong Sun, Yunfeng Zhu, Xiaofang Zhang, Zhenyu Chen. Proceedings of the 34th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), 2025: 1701–1724.
 4. [Co-First Author | ACL Findings 2026 | CCF A, Core Ranking A*]
[Train in Vain: Functionality-Preserving Poisoning to Prevent Unauthorized Use of Code Datasets.](#)
Yuan Xiao*, Jiaming Wang*, Yuchen Chen*, Wei Song, Jun Sun, Shiqing Ma, Yanzhou Mu, Juan Zhai, Chunrong Fang, Jin Song Dong, Zhenyu Chen. Findings of the Association for Computational Linguistics: ACL 2026, 2026: 31284–31303.
 5. [Co-First Author | FSE 2026 | CCF A, Core Ranking A*]
[DuCodeMark: Dual-Purpose Code Dataset Watermarking via Style-Aware Watermark-Poison Design.](#)
Yuchen Chen*, Yuan Xiao*, Chunrong Fang, Zhenyu Chen, Baowen Xu. Proceedings of the ACM on Software Engineering, ACM International Conference on the Foundations of Software Engineering (FSE), 2026: 1–23.
 6. [Co-Author | ISSTA 2024 | CCF A, Core Ranking A]
[CooTest: An Automated Testing Approach for V2X Communication Systems.](#)
An Guo, Xinyu Gao, Zhenyu Chen, Yuan Xiao, Jiakai Liu, Xiuting Ge, Weisong Sun, Chunrong Fang. Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), 2024: 1453–1465.
 7. [Co-Author | FSE 2025 | CCF A, Core Ranking A*]
[Eliminating Backdoors in Neural Code Models for Secure Code Understanding.](#)
Weisong Sun, Yuchen Chen, Chunrong Fang, Yebo Feng, Yuan Xiao, An Guo, Quanjuan Zhang, Zhenyu Chen, Baowen Xu, Yang Liu. Proceedings of the ACM on Software Engineering, ACM International Conference on the Foundations of Software Engineering (FSE), 2025: 1386–1408.
 8. [Co-Author | TSE 2026 | CCF A Journal]
[Securing Code Understanding: Detecting Natural Backdoor Vulnerability in Code Language Models.](#)
Yuchen Chen, Weisong Sun, Haocheng Huang, Yuan Xiao, Chunrong Fang, Yiran Zhang, Tingting Xu, Zhenpeng Chen, An Guo, Peizhuo Lv, Xiaofang Zhang, Zhenyu Chen, Yang Liu, Baowen Xu. IEEE Transactions on Software Engineering, 2026: 1–18.
 9. [Co-Author | ISSTA 2026 | CCF A, Core Ranking A]
[Security Risks for LLM-based Code Generation.](#)
Yuchen Chen, Wei Cheng, Yuan Xiao, Zhou Yang, Weifeng Sun, Chunrong Fang, Xiang Chen, Baowen Xu, David Lo, Zhenyu Chen. ACM International Symposium on Software Testing and Analysis (ISSTA), 2026.
 10. [Co-Author | ASE 2026 | CCF A, Core Ranking A*]
[Breaking Customized LLMs for Coding: Automated Red Teaming for Instruction Backdoor Attacks.](#)
Yuchen Chen, Wei Cheng, Yuan Xiao, Weisong Sun, Chunrong Fang, Yang Liu, Zhenyu Chen, Baowen Xu. Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (ASE), 2026.
 11. [Co-Author | FSE 2026 | CCF A, Core Ranking A*]
[PuzzleMark: Implicit Jigsaw Learning for Robust Code Dataset Watermarking in Neural Code Completion Models.](#)
Haocheng Huang, Yuchen Chen, Weisong Sun, Peizhuo Lv, Yuan Xiao, Chunrong Fang, Yang Liu, Xiaofang Zhang. ACM International Conference on the Foundations of Software Engineering (FSE), 2026.

Recent Progress (Core research completed; manuscripts in preparation)

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1. [First Author]
MPW: Statistically Stealthy Code Dataset Watermarking for Training Data Ownership Verification.
Yuan Xiao et al.
 2. [First Author]
Persistence and Detectability of Code Dataset Watermarks under Model Evolution.
Yuan Xiao et al.
 3. [First Author]
White-Box Attribution for Verifying Training Data Usage in Code Models.
Yuan Xiao et al.

Selected Talks & Presentations

1. “Code Dataset Watermarking for Code Large Language Models.”
Invited Talk, Postgraduate Forum on Large-Model Data Governance and Efficient Training, Harbin Institute of Technology, Shenzhen, Dec. 2025.
2. “Code Dataset Watermarking for Code Large Language Models.”
Research Presentation, The 16th International Young Scholars Forum of Sun Yat-sen University, Shenzhen and Zhuhai, China, Dec. 2025.
3. “DeCoMa: Detecting and Purifying Code Dataset Watermarks through Dual Channel Code Abstraction.”
Conference Presentation, ISSTA 2025.
4. “Towards General Robustness Verification of MaxPool-Based Convolutional Neural Networks via Tightening Linear Approximation.”
Poster Presentation, CVPR 2024.

Academic Service (Reviewer / Co-Reviewer)

- Reviewer: AAAI 2023; NeurIPS 2024, 2025, 2026; ICLR 2025, 2026; ICML 2025
- Co-Reviewer: AAAI 2025; USENIX 2026; TASE 2025
- Teaching Assistant, “Foundations of Data Science,” 2021–2024

Honors

Outstanding Ph.D. Graduate	2026
<i>Nanjing University</i>	<i>Top $\leq 15\%$</i>
Huawei Scholarship (Doctoral)	2025
<i>Nanjing University</i>	<i>2 recipients, School of Software</i>
Outstanding Graduate Student Exemplar	2025
<i>Nanjing University</i>	<i>Top $\leq 1.5\%$</i>
Outstanding Graduate Student	2022
<i>Nanjing University</i>	
Graduate Academic Scholarship	2021 – 2025
<i>Nanjing University</i>	
University Scholarship, 2nd and 3rd Prize	2017 – 2021
<i>Sun Yat-sen University</i>	
National 2nd Prize, National Undergraduate Mathematical Contest in Modeling	2019
<i>National Level</i>	<i>Top 3.1% nationally</i>